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SPECIFICATION FOR APPROVAL

DATE:

CUSTOMER: _____

PART NAME: Metal Strip Type Halogen Free Current Sensing Resistors

CUSTOMER'S DWG. NO. _____

CUSTOMER'S PART NO. _____

PDC PART NO. FMF SERIES APPROVED

DESCRIPTION. _____

RESULT / ACTION	" ˇ "	CUSTOMER'S SIGNATURE	NOTE
FULL APPROVED			
CONDITIONAL APPROVED			
REJECTED			

OUR ACTION	SIGNATURE
PREPARED BY	*Jenny Tseng*
CHECKED BY	*Tony Chou*
APPROVED BY	*Byron Tsai*

CUSTOMER SIGNATURE FOR ACCEPTANCE

FMF series
Metal Strip Type
Lead Free Current Sensing Resistors

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FMF series
Metal Strip Type
Lead Free Current Sensing Resistors

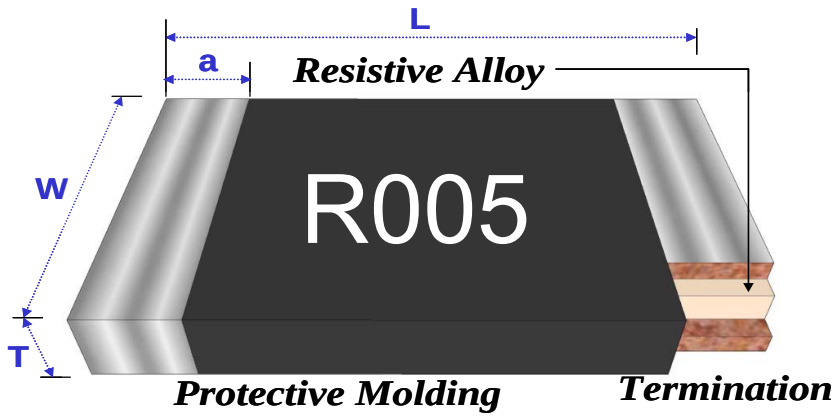
1. Features

- High power rating and low TCR.
- Low resistance and high precision (1%).
- Inductance less than 1.0nH.
- Excellent reliability and suitable cost.
- Suitable for lead free soldering.
- RoHS compliant & Halogen Free

2.Applications

- Switching model power supply.
- Battery pack.
- Notebook, Tablet PC
- Test Instrument.
- Power Amplifier.

3.Dimension and Construction



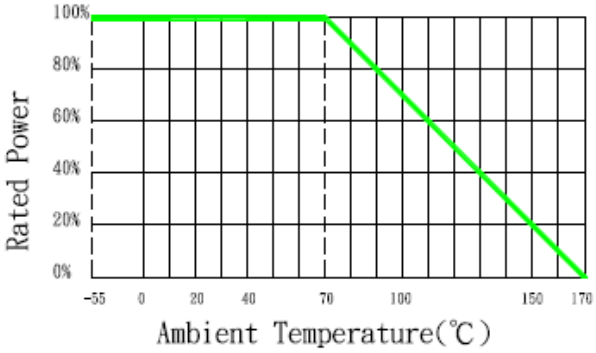
Item	Protective Molding	Resistive Element	Internal Terminal	External Terminal
Material	Resin	Alloy Metal	Copper	Solder

Unit：mm

Type	L	W	T	a
FMF05 5~20m	2.00±0.15	1.20±0.15	0.45±0.15	0.325±0.20
FMF06 5m~30m	3.10±0.20	1.65±0.20	0.60±0.20	0.60±0.20
FMF25 3m~100m	6.20±0.20	3.25±0.20	0.60±0.20	0.80±0.20
FMF59 2~10m	15.0±0.20	7.80±0.20	0.70±0.20	3.50±0.20

4.Power Derating Curve

Operating Temperature Range: -55 to +170 deg.C



5.Rating

Rating / Type	Power Rating at 70℃(W)	Max Working Voltage	Max Overload Voltage	Resistance Tolerance (%)	Temperature coefficient of Resistance (ppm/℃)	Resistance (mΩ)
FMF05 0805	0.25	71mV	158mV	±1%(F) ±2%(G) ±5%(J)	±100	5,6,7,8,9,10,20
	0.5	100mV	224mV			
FMF06 1206	0.5	123mV	274mV	±1%(F) ±2%(G) ±5%(J)	±70	5,10,15,20,25,30
	1	173mV	387mV		±70	5,10,15,20,25,30
FMF25 2512	1	316mV	707mV	±1%(F) ±2%(G) ±5%(J)	±100	3,4
					±70	5,6,7,8,9,10,12,15, 20,25,50,100
	2	447mV	1000mV		±100	3,4
					±70	5,6,7,8,9,10,12,15, 20,25,30,33,35,40, 50,100
FMF59 5931	5	224mV	500mV	±1%(F) ±2%(G) ±5%(J)	±100	2,3
					±70	5,10

Note:

(i) E = $\sqrt{P*R}$ or Max. Working Voltage whichever is lower.

(ii) E：Working Voltage(V)，P：Rated Power (W)，R：Resistance Value(Ω)

(iii) Solder-pad and trace size should be >300 mm^2 and board surface temperature should not exceed 105℃ when applying rated power

6.Part Number

Type	Size	Tolerance	Packing	Watt	R Value	TCR	Special Code
FMF	**05** :0805 **06** :1206 **25** :2512 **59** :5931	**F** :±1% **G** :±2% **J** :±5%	**T** :Paper Tape 5Kpcs (For 0805) 4Kpcs (For 1206) **P** :Plastic Tape 4Kpcs (For 2512) **Q** :Plastic Tape 3Kpcs (For 5931)	**D**: 1/4W **F**: 1/2W **H**: 1W **J**: 2W **M**: 5W	**RXXX** 4 digit	**-**: As Rating Info	2512/1206 5931： **L** ⇔ **LH**: Standard **B** ⇔ **BH**: Low EMF
							0805： **LH** : Standard **BH** : Low EMF
							All size： AEC-Q200 **LHM** : Standard **BHM** : Low EMF

Example：

FMF25FPJR005-BH ⇔ FMF25FPJR005-B

➔Metal strip, 2512 size, ±1%, plastic tape, 2W, 5mΩ, low emf

FMF25FPJR010-BHM

➔Metal strip, 2512 size, ±1%, plastic tape, 2W, 10mΩ, low emf, AEC-Q200

FMF05FTFR010-LHM

➔Metal strip, 0805 size, ±1%, paper tape, 1/2W, 10mΩ, standard, AEC-Q200

7.Marking

FMF 1206/2512/5931

TOP：Marking. (4 Digits marking to identify the resistance value.)



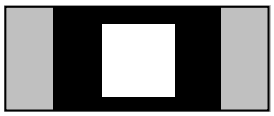
R005=5mΩ，R020=20mΩ

FMF 0805

TOP：No marking product.



Bottom：Index sign.



8. Reliability Performance (AEC-Q200)

***** Normal test items for standard product.

Test Item	Specification	Test Method (AEC-Q200. IEC 60115)
***DC Resistance**	F：±1%、G：±2%、J：±5%	**AEC-Q200 TABLE 7.1** **IEC 60115-1 / JIS C 5201-1 , Clause 4.5** Measure the resistance Value.
High Temperature Exposure (Storage)	J、G：△R≦ ±(3%+0.5mΩ) F：△R≦ ±(1%+0.5mΩ)	**AEC-Q200 TABLE 7.3** 1000 hrs. @ T=125℃. Unpowered. Measurement at 24 ±2 hours after test conclusion.
***Temperature Cycling**	J、G：ΔR≦ ±(1%＋1mΩ) F：ΔR≦ ±(0.5%＋1mΩ) No mechanical damage.	**AEC-Q200 TABLE 7.4** 1000 Cycles (-55°C to +125°C). Measurement at 24±2 hours after test conclusion.
Moisture Resistance	J、G：△R≦ ±(1%+0.5mΩ) F：△R≦ ±(0.5%+0.5mΩ)	**AEC-Q200 TABLE 7.6** Test 65℃/80~100%RH/10Cycles. Measurement at 24±2 hours after test conclusion. (t=24hrs/cycle).

Biased Humidity	J、G：△R≦ ±(3%+0.5mΩ) F：△R≦ ±(1%+0.5mΩ)	**AEC-Q200 TABLE 7.7** 1000 hours 85℃/85%RH. 10% of operating power. Measurement at 24 ±2 hours after test conclusion.
Operational Life	J、G：△R≦ ±(3%+0.5mΩ) F：△R≦ ±(1%+0.5mΩ)	**AEC-Q200 TABLE 7.8** Test 1000hr @ TA=125℃ at specified rated power. Measurement at 24±2 hours after test conclusion.
External Visual	No visual damage and refer PDC marking code.	**AEC-Q200 TABLE 7.9** Inspect device construction, marking and workmanship.
Physical Dimension	Within the spec.	**AEC-Q200 TABLE 7.10** Verify physical dimensions to the applicable device detail specification.
Mechanical Shock	Within product specification tolerance and no visible damage.	**AEC-Q200 TABLE 7.13** Test Peak value:100g's,Wave:Hail-sine, Duration:6ms,Velocity:12.3ft/sec.
Vibration	No mechanical damage.	**AEC-Q200 TABLE 7.14** 5 g's for 20 min., 12 cycles each of 3 orientations. Test from 10-2000 Hz.
***Resistance to Solder Heat**	J、G：ΔR≦ ±(1%＋0.5mΩ) F：ΔR≦ ±(0.5%＋0.5mΩ) No mechanical damage.	**AEC-Q200 TABLE 7.15** Solder dipping @ 270℃±5℃ for 10sec.±1sec.
Thermal Shock	J、G：ΔR≦ ±(1%＋0.5mΩ) F：ΔR≦ ±(0.5%＋0.5mΩ) No mechanical damage.	**AEC-Q200 TABLE 7.16** -55 to 155℃/ dwell time 15min/ Max transfer time 20sec/ 300cycles.
ESD	ΔR≦ ±(1%＋0.5mΩ) No mechanical damage.	**AEC-Q200-002** Test contact min. 1KV.
***Solder Ability**	Over 95% of termination must be covered with solder.	**AEC-Q200 TABLE 7.18** a)Baking 155℃ 4H, dipping 235℃ 5s b)Steam 1H, dipping 215℃ 5s c)Steam 1H, dipping 260℃ 7s

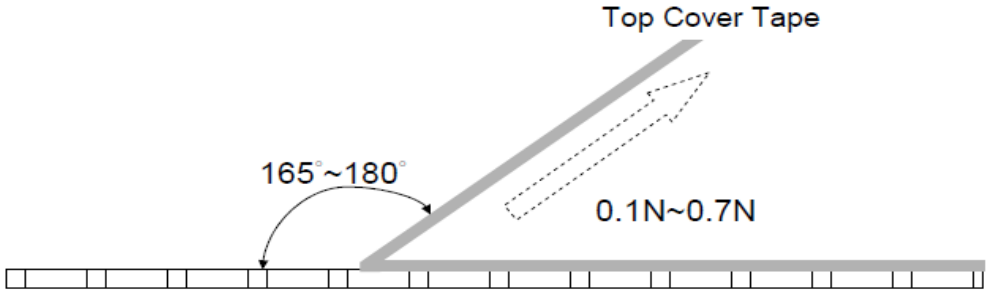
Flammability	Refer UL-94.	**AEC-Q200 TABLE 7.20** UL-94 V-0 or V-1 are acceptable
***Board Flex**	J、G：ΔR≦ ±(1%＋1mΩ) F：ΔR≦ ±(0.5%＋1mΩ) No mechanical damage.	**AEC-Q200 TABLE 7.21** Bending 2mm 5931.2512.2010.1210.1206 3mm 0805
Terminal Strength	No mechanical damage	**AEC-Q200 TABLE 7.22** Force 1 Kg for 60 seconds.
***Short Time Overload**	J、G：ΔR≦ ±(2%＋0.5mΩ) F：ΔR≦ ±(1%＋0.5mΩ)	**IEC 60115-1, Clause 4.13** 5 × Rated power for 5 seconds
***Load Life Humidity**	J、G：△R≦ ±(3%+0.5mΩ) F：△R≦ ±(1%+0.5mΩ)	**IEC 60115-1, Clause 4.24** 40±2℃ with relative humidity 90% ~ 95% D.C. rated voltage for 1.5 hours ON 30 minutes OFF. Cycle repeated 1000 hours.
***Temperature Coefficient of Resistance (TCR)**	Within the spec.	**IEC 60115-1, Clause 4.8** T_1 T_2 Test temperature： 25℃~ -55℃ 25℃~ +155℃ TCR(ppm/℃) = $(R_2-R_1)/R_1 \times 1 / (T_2-T_1) \times 10^6$
***Load Life**	J、G：△R≦ ±(3%+0.5mΩ) F：△R≦ ±(1%+0.5mΩ)	**IEC 60115-1, Clause 4.25** Rated voltage for 1.5 hours for followed by a pause 0.5 hour at 70±2℃. Cycle repeated 1000 hours.
***Insulation Resistance**	Between termination and coating must over 1000MΩ	**IEC 60115-1, Clause 4.6** Test voltage：100±15V

9. PACKAGING

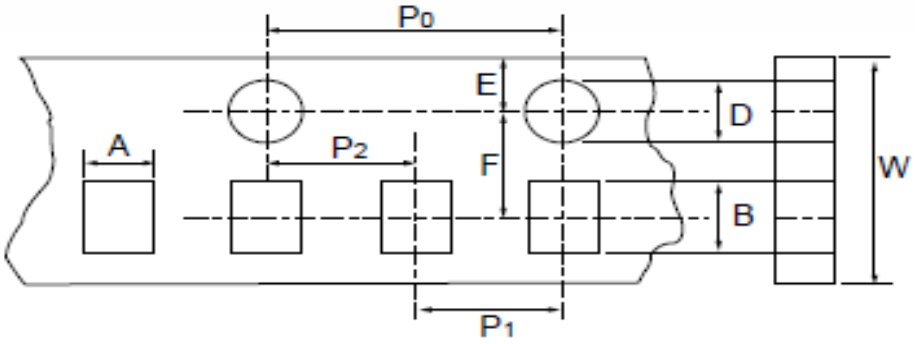
9.1 Peel Strength of Top Cover Tape

The peel speed shall be about 300 mm/min

The peel force of top cover tape shall between 0.1 to 0.7N



9.2 Tape Packaging Dimensions

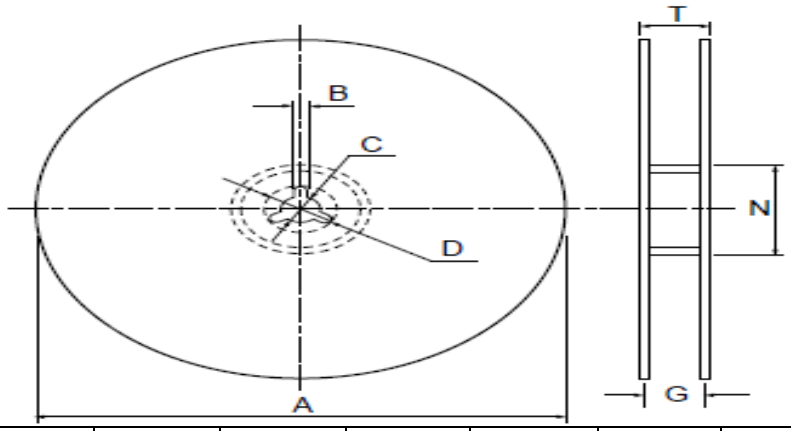


- Accumulated dimensional tolerance 40±0.2mm

Size	A	B	W	F	E	P1	P2	P0	D
0805	1.65±0.20	2.40±0.20	8.00±0.30	3.50±0.05	1.75±0.10	4.00±0.10	2.00±0.05	4.00±0.10	1.50±0.10
1206	2.00±0.20	3.60±0.20	8.00±0.30	3.50±0.05	1.75±0.10	4.00±0.10	2.00±0.05	4.00±0.10	1.50±0.10
2512	3.50±0.20	6.75±0.20	12.0±0.30	5.50±0.05	1.75±0.10	4.00±0.10	2.00±0.05	4.00±0.10	1.50±0.10
5931	8.20±0.10	15.4±0.10	24.0±0.30	11.5±0.10	1.75±0.10	12±0.10	2.00±0.10	4.00±0.10	1.50±0.10

unit : mm

9.3 Reel Dimensions



Size	Packaging Q'ty	A	N	C	D	B	G	T
0805	5kpcs/Reel	178.0±2.0	60.0±0.5	13.0±0.5	20(Min.)	2.0±0.5	10.0±1.5	14.9max.
	10kpcs/Reel	254.0±2.0	100.0±1.0	13.5±0.5	20(Min.)	2.0±0.5	10.0±1.5	14.9max.
	20kpcs/Reel	330.0±2.0	100.0±1.0	13.5±0.5	20(Min.)	2.0±0.5	10.0±1.5	14.9max.
1206	4kpcs/Reel	178.0±2.0	60.0±0.5	13.0±0.5	20(Min.)	2.0±0.5	10.0±1.5	14.9max.
2512	4kpcs/Reel	178.0±2.0	60.0±0.5	13.0±0.5	20(Min.)	2.0±0.5	13.8±1.5	16.7max.
	8kpcs/Reel	254.0±2.0	100.0±0.5	13.5±0.5	20(Min.)	2.0±0.5	13.8±1.5	20.0max.
	16kpcs/Reel	330.0±2.0	100.0±1.0	13.5±0.5	20(Min.)	2.0±0.5	13.8±1.5	20.0max.
5931	3kpcs/Reel	330.0±2.0	99.5±1.0	13.0±0.5	20(Min.)	2.0±0.5	24.0±1.0	28.6max.

10. Storage &. Handling

… Products are recommended to be used up within one year as ensured shelf life.
Check solder ability in case shelf life extension is needed.

… To store products with following condition:
Temperature:5 to 40℃；Humidity: 20 to 70% relative humidity.

Precaution for use：

The AEC-Q200 series resistors is mainly used on general automotive equipment without safety considerations. Please contact our company in advanced if you intend to use resistor for designing the equipment which may damage itself and the safety of third party. If necessary, please consider to add the protect circuit in devising process and obtaining fully safety evaluation. The contents of the acknowledgment is only used for our parent company, marketing subsidiaries and official marketing agents who purchase our products. Not applicable for the other nonofficial channels.

※ All product specification and data are subject to change without notice.